

The θ Threshold and Bayesian Probability

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Main idea: first, we define what would make a process problematic. Next, we use data to calculate how likely it is that the problem exists. Finally, we consider the costs of possible errors to decide whether we should act.

The objective is to distinguish three quantities:

$$\theta_0 \quad q = P(\theta > \theta_0 \mid D) \quad \tau$$

1 The true rate and its threshold

Imagine a food manufacturing plant. We will use θ to represent the true proportion of contaminated items produced by the line:

θ = true contamination rate.

If $\theta = 0.10$, approximately 10% of the items would be contaminated in the long run. The value of θ is unknown because we cannot test every item produced in the past or every item that will be produced in the future.

Management considers a rate above 25% unacceptable. This threshold is

$$\theta_0 = 0.25.$$

The adverse state is

$$\theta > \theta_0,$$

or, in this example,

$$\theta > 0.25.$$

The number 0.25 defines when the true rate is considered too high. It does not yet tell us the probability that the problem exists.

1.1 An analogy

A speed limit of 80 miles per hour defines what counts as speeding, but it does not tell us the probability that a car is traveling above the limit. Similarly, $\theta_0 = 0.25$ defines what counts as excessive contamination, but it does not tell us the probability that θ is above that value.

2 The prior, the data, and the posterior

Before testing,

$$\theta \sim \text{Beta}(2, 8).$$

The plant tests 20 items and finds 8 contaminated items. There are 8 positive results and 12 negative results. Using the Beta–Binomial update,

$$\theta \mid D \sim \text{Beta}(2 + 8, 8 + 12) = \text{Beta}(10, 20).$$

The symbol D represents the observed data. The expression $\theta \mid D$ means the distribution of θ after considering the data.

The observed proportion is

$$\frac{8}{20} = 0.40,$$

while the posterior mean is

$$E(\theta \mid D) = \frac{10}{10 + 20} = 0.3333.$$

The posterior mean of 0.3333 does not directly answer the question of how likely θ is to be greater than 0.25. To answer that question, we must consider the entire posterior distribution.

3 The Bayesian probability q

Because θ is unknown, we calculate

$$q = P(\theta > 0.25 \mid D).$$

This is the probability that the true contamination rate is greater than 25%, given the observed data.

3.1 Graph of the posterior distribution

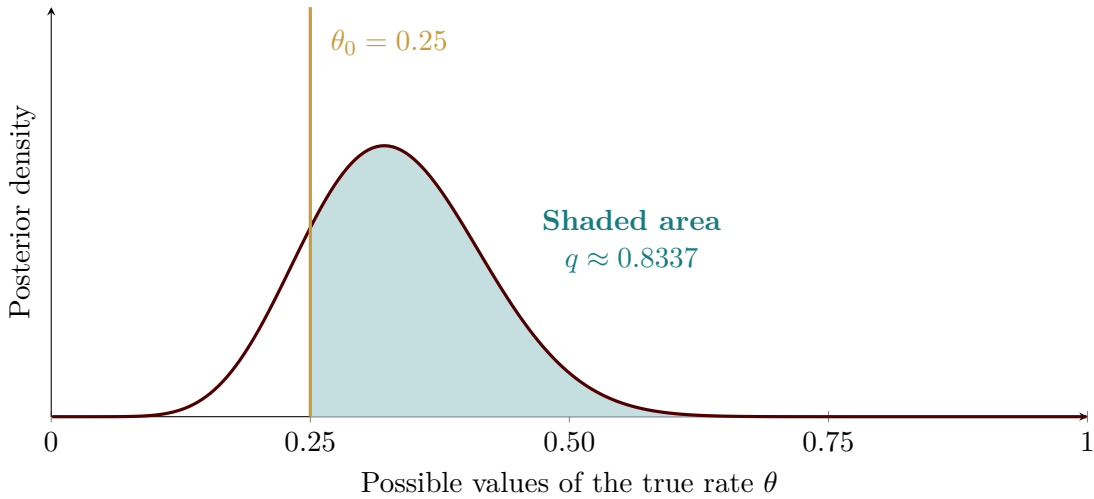


Figure 1: Posterior Beta(10, 20) distribution. The shaded region represents $P(\theta > 0.25 \mid D)$.

The shaded area is

$$q = P(\theta > 0.25 \mid D) \approx 0.8337.$$

Given the model, the prior, and the test results, we assign an 83.37% posterior probability to the claim that the true contamination rate exceeds 25%.

4 How to calculate q

The cumulative distribution function gives the probability that θ is to the left of a specified value:

$$F_{\text{Beta}(10,20)}(0.25) = P(\theta \leq 0.25 \mid D) \approx 0.166305.$$

Because we want the area to the right,

$$P(\theta > 0.25 \mid D) = 1 - P(\theta \leq 0.25 \mid D).$$

Therefore,

$$q = 1 - 0.166305 = 0.833695 \approx 0.8337.$$

4.1 Calculation in R

```
pbeta(  
  0.25,  
  shape1 = 10,  
  shape2 = 20,  
  lower.tail = FALSE  
)
```

4.2 Calculation in Python

```
from scipy.stats import beta  
beta.sf(0.25, a=10, b=20)
```

In both cases, the approximate result is 0.8336951.

5 Why θ_0 and q are different

Quantity	Value	Meaning
θ_0	0.25	Threshold that defines an unacceptable rate
q	0.8337	Probability that θ is above θ_0

First, we use θ_0 to construct the event. Next, we calculate the posterior probability of that event:

$$\theta_0 = 0.25 \longrightarrow \theta > 0.25 \longrightarrow q = P(\theta > 0.25 \mid D).$$

Comparing 0.8337 directly with 0.25 is not the decision rule. To choose an action, we must compare q with the action threshold τ .

6 The decision threshold τ

The decision depends on the consequences of two types of error.

Decision	Rate is truly high	Rate is not high
Stop and sanitize	Correct decision	False positive: unnecessary shutdown
Continue production	False negative: untreated problem	Correct decision

Suppose that

$$C_{\text{FP}} = 40,000 \quad \text{and} \quad C_{\text{FN}} = 120,000.$$

The decision threshold is

$$\tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}} = \frac{40,000}{40,000 + 120,000} = 0.25.$$

We now compare two probabilities:

$$q = 0.8337 > \tau = 0.25.$$

Because $q > \tau$, the Bayesian decision is to **stop and sanitize the production line**.

7 The coincidence of the two values equal to 0.25

In this example, θ_0 and τ are both equal to 0.25, but this is only a numerical coincidence:

$\theta_0 = 0.25$: When is the process rate considered excessive?

$\tau = 0.25$: How much posterior probability is required to act?

If the two types of error had equal costs,

$$\tau = \frac{1}{1 + 1} = 0.50,$$

but the adverse state would still be defined as $\theta > 0.25$.

If management changed the adverse state to $\theta > 0.35$, then

$$q_{0.35} = P(\theta > 0.35 \mid D) \approx 0.4076.$$

Because the costs did not change, τ would remain equal to 0.25. Since

$$0.4076 > 0.25,$$

the decision would still be to stop and sanitize.

Changing θ_0 changes the event whose probability we calculate. Changing the costs changes τ . Neither change, by itself, changes the Beta(10, 20) posterior distribution.

8 Summary of the reasoning

1. Define the state:

$$\theta > \theta_0.$$

2. Update the distribution:

$$\theta \mid D \sim \text{Beta}(10, 20).$$

3. Calculate the probability:

$$q = P(\theta > \theta_0 \mid D).$$

4. Convert the costs into a threshold:

$$\tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}}.$$

5. Compare:

if $q > \tau$, choose the positive action.

Remember: θ_0 defines the problem, q measures our confidence that the problem exists, and τ determines when it is worthwhile to act.

9 Check your understanding

1. What does $\theta_0 = 0.25$ represent?
2. What does $q = 0.8337$ mean?
3. Why do q and θ_0 not play the same role?
4. With which quantity should q be compared when choosing an action?

Answers

1. It is the threshold that determines when the true rate is excessive.
2. It is the posterior probability that θ exceeds 0.25.
3. Because θ_0 defines an event, while q measures the probability of that event.
4. With τ , the threshold derived from the costs.